

R2_Score for Profit Prediction

Multiple Linear Regression – R2_score=0.93586

Support Vector Machine Regression

S.No	Hyper Parameter	linear (r2_score)	Rbf (r2_score)	Poly (r2_score)	Sigmoid (r2_score)
1	C=10	-0.0396	-0.05680	-0.05366	-0.05471
2	C=100	0.10646	-0.05072	-0.01980	-0.03045
3	C=500	0.59289	-0.02432	0.11468	0.07057
4	C=1000	0.78028	0.00067	0.26616	0.18506
5	C=2000	0.87677	0.06751	0.48100	0.39706
6	C=3000	0.89567	0.12322	0.63700	0.59136
7	C=5000	0.90037	0.21242	0.79365	0.73065
8	C=7000	0.91819	0.27880	0.82987	0.82596

SVM Regression– R2_score=0.91819 (C=7000, kernel='linear')

Decision Tree Regression

S. No	Criterion	Splitter	R2_Score
1	<i>squared_error</i>	best	0.92228
2	<i>squared_error</i>	random	0.90846
3	<i>friedman_mse</i>	best	0.92497
4	<i>friedman_mse</i>	random	0.89574
5	absolute_error	best	0.95221
6	<i>absolute_error</i>	random	0.90522
7	<i>poisson</i>	best	0.92093
8	<i>poisson</i>	random	0.74698

Decision Tree Regression - R2_score=0.95221 (criterion='absolute_error', splitter='best')